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Complete reporting of clinical trials requires more than journal articles

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First published in 1996, CONSORT (Consolidated Standards of Reporting Trials) emphasised the importance of accurate and complete reporting of randomised clinical trials.¹ Journal articles were the only public records of many randomised clinical trials, and the CONSORT reporting guidelines described the minimum information those articles should include. Most recommendations focused on methods and results. At the time, articles were published on paper and supplements were limited. Infrastructure to register clinical trials and to share other artefacts (eg, data, code) had not been invented. The first CONSORT update in 2001 described trial registration as desirable, but not essential.² When a CONSORT update in 2010 added trial registration as a checklist item,³ opportunities for sharing data and code were still relatively new.⁴ SPIRIT (Standard Protocol Items: Recommendations for Interventional Trials) 2013, the first reporting guidelines for trial protocols, introduced recommendations about data and code sharing.⁵

CONSORT 2025 (doi:10.1136/bmj-2024-081123) and the soon-to-be published SPIRIT 2025 (doi:10.1136/bmj-2024-081477) update and replace previous versions of these essential reporting guidelines.^{6,7} These updates are the first since guidance for statistical analysis plans was published in 2017⁸ and the first since data and code sharing became relatively commonplace. Updating both guidelines in tandem has led to helpful clarifications and harmonisation. The updated guidelines also include some new items. An especially welcome section on “Open science” recommends reporting whether other research artefacts are publicly available. This new section highlights an important shift from focusing on journal articles towards a contemporary understanding of “reporting” as sharing a collection of research artefacts associated with a study.

The CONSORT 1996 checklist fit on half a page. The developers predicted the checklist would grow as researchers identified more information considered important for all randomised clinical trials and for specific types of these trials.¹ Indeed, the detailed SPIRIT 2025 and CONSORT 2025 checklists are 24 and 12 pages long, respectively.^{9,10} Consequently, it might be difficult to follow the CONSORT 2025 guidance that every checklist item be reported “somewhere in the article, with sufficient detail and clarity.” Additionally, the Enhancing the Quality and Transparency Of health Research (EQUATOR) Network library—an online repository of reporting guidelines (<https://www.equator-network.org>)—identifies 32 CONSORT and 10 SPIRIT extensions for different types of research questions (eg, dose finding^{11,12}), frameworks (eg, non-inferiority¹³), designs (eg, cluster randomised,¹⁴

n-of-1 trials^{15,16}), interventions (eg, social and psychological,¹⁷ herbal¹⁸), outcomes (eg, harms,¹⁹ surrogates,^{20,21} patient reported^{22,23}), and article sections (eg, abstracts,²⁴ intervention descriptions,²⁵ outcomes^{26,27}). Editors had already begun to complain in 2015 when 10 CONSORT extensions had been produced.²⁸ Multiple extensions now apply to most randomised clinical trials, so users are typically advised to consult four or five documents.²⁹

The current system is more comprehensive than CONSORT 1996, but it is not always easy to use. Notably, it is no longer clear where and how all the recommended items should be reported so that every reader can find the information that matters to them. While CONSORT 1996 focused on the main text of journal articles—even recommending the organisation of information under five suggested subheadings—CONSORT 2025 stops short of stating which items are essential for the main text and which might be included in supplementary materials. Consequently, it might be difficult for authors, peer reviewers, and editors to implement its recommendations. Clearer guidance is needed about what information is essential for the main text of every journal article and where the other recommended information belongs.

There is limited evidence that reporting guideline endorsement is associated with more complete reporting,³⁰ and the developers of SPIRIT 2025 and CONSORT 2025 rightly argue that reporting quality remains suboptimal.³¹ Standards for assessing adherence to SPIRIT 2025 and CONSORT 2025 could help bridge the gap between recommendations and practice. For some items, the presence or absence of information might be easy to assess (eg, allocation ratio). For other items, it is unclear what constitutes complete reporting (eg, background and rationale, trial settings). The developers say that SPIRIT 2025 and CONSORT 2025 are not tools for assessing trial quality (eg, risk of bias), but many studies have and will use reporting guidelines to assess reporting quality, including important studies by the developers.^{32–37} Moreover, SPIRIT and CONSORT developers have rightly argued that reporting guidelines should be used by peer reviewers and editors to improve reporting quality.^{38,39} Objective criteria for assessing adherence to each item could help users understand, assess, and implement recommendations. Rules would also support the use of emerging technologies to facilitate manuscript writing and review, such as artificial intelligence.

It might not be feasible for a few thousand words in a journal article to constitute a complete report of a randomised clinical trial. However, infrastructure and technology better support research transparency today than in 1996. Expanding the scope of reporting guidelines to include recommendations for sharing

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and organising multiple research artefacts would promote greater transparency and openness. For example, the Transparency and Openness Promotion (TOP) guidelines⁴⁰ provide a framework that journals and others can use to promote research transparency. TOP describes seven practices that support verification of study results (eg, sharing study protocols, statistical analysis plans, research materials, analytical code, and data). Future updates to SPIRIT and CONSORT could articulate how those practices should be implemented to make randomised clinical trials verifiable and useful for multiple interest holders. For example, it might be unrealistic to include all items about statistical methods in the main text, but it should be possible to understand and audit analyses using freely available and permanent copies of statistical analysis plans and code. Future updates might state that journal articles should include links to intervention protocols. Within ethical and legal limits, individual data should be available to support reanalysis, new research, and evidence synthesis. A truly complete report of a randomised clinical trial would include all these things in addition to a journal article summarising the methods and results.

The new “Open science” section underscores that reporting a randomised clinical trial is not synonymous with publishing a journal article. SPIRIT, CONSORT, and other reporting guidelines should continue to develop recommendations about the collection of artefacts that are essential for different study types. SPIRIT and CONSORT are guidelines for randomised clinical trials, so updates and extensions should describe the minimum information to be included in the main text of articles summarising randomised clinical trials, other artefacts that should be available for every randomised clinical trial, and what information those artefacts should contain to meet the needs of various interest holders. To improve implementation and research, objective standards for assessing adherence to each item in these reporting guidelines should be developed.

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